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What we proved for $\mathbb{Z}[\sqrt{d}]$ last time:

Assuming as always $d \in \mathbb{Z}$ is not a square.

1. $\alpha = 0 \Leftrightarrow N(\alpha) = 0$

2. $\overline{\alpha\beta} = \bar{\alpha} \bar{\beta}$.

$$N(\alpha\beta) = N(\alpha)N(\beta).$$

$$\alpha \mid \beta, \beta \neq 0 \Rightarrow |N(\alpha)| \leq |N(\beta)|.$$

3. α is a unit $\Leftrightarrow |N(\alpha)| = 1$.

4. $|N(\alpha)|$ is a prime number $\Rightarrow \alpha$ is prime.

Existence of prime factorizations in $\mathbb{Z}[\sqrt{d}]$.

Lem. Every nonzero, nonunit $\alpha \in \mathbb{Z}[\sqrt{d}]$ is divisible by a prime.

pf. α is prime. $\checkmark$

α is composite: α is divisible by some nonunit, non-

associate β . Let π be such a divisor with minimal

$$|N(\pi)| > 1.$$

Claim: π is prime.

Let γ/π be nonunit. Then $|N(\gamma)| \mid |N(\pi)|$. So $|N(\gamma)| \leq$

$|N(\pi)|$. Since γ is nonunit and γ/π , $\pi/\alpha \Rightarrow \gamma/\alpha$.

by minimality of $|N(\pi)|$ we have $|N(r)| = |N(\pi)|$. Thus

r is an associate of π . Hence π is prime. $\square$

Thm (Existence of prime factorizations in $\mathbb{Z}[\sqrt{d}]$) Assume $d \in \mathbb{Z}$

is not a square. Every nonzero, nonunit element can be

written as a (finite) product of primes.

Pf. Let $\alpha \in \mathbb{Z}[\sqrt{d}]$ be nonzero, nonunit.

α is prime $\checkmark$

α is composite: Assume to the contrary that α is

an element with no prime factorizations and minimal

$|N(\alpha)| > 1$. By Lem. there is a prime $\pi_1 | \alpha$. Suppose

$\alpha = \pi_1 \beta$ for some $\beta \in \mathbb{Z}[\sqrt{d}]$. Since $|N(\pi_1)| > 1$ and

$N(\alpha) = N(\pi_1) N(\beta)$, we see

$$1 \leq |N(\beta)| = \frac{|N(\alpha)|}{|N(\pi_1)|} < |N(\alpha)|.$$

Because α is composite, β is nonzero. By minimality of

$|N(\alpha)|$, β has a prime factorization $\beta = \pi_2 \pi_3 \cdots \pi_k$.

Consequently, $\alpha = \pi_1 \beta = \pi_1 \pi_2 \cdots \pi_k$ is a prime factorization

of α . This is a contradiction. $\square$

Q! Is every $\mathbb{Z}[\sqrt{d}]$ a UFD: every nonzero, nonunit element in $\mathbb{Z}[\sqrt{d}]$

can be written as a product of primes unique up to order and

multiplication by units?

Examples of nonunique factorizations

1. $\mathbb{Z}[\sqrt{-3}]$ ($d = -3$): $4 = 2 \cdot 2 = (1 + \sqrt{-3})(1 - \sqrt{-3})$

(1) 2 is prime in $\mathbb{Z}[\sqrt{-3}]$.

Let $\alpha \mid 2$. Then $|N(\alpha)| \mid |N(2)| = 2^2 = 4$. So $|N(\alpha)| = 1, 2, 4$.

① $|N(\alpha)| = 1$; α is unit.

② $|N(\alpha)| = 4$: $|N(\alpha)| = |N(2)|$. So α is an associate of 2.

③ $|N(\alpha)| = 2$: Let $\alpha = x + y\sqrt{3}$. Then

$$|N(\alpha)| = |(x + y\sqrt{3})(x - y\sqrt{3})| = x^2 - 3y^2 = 2.$$

$$\text{Mod } 3: \quad x^2 \equiv 2 \pmod{3}.$$

But this is impossible, because $2 \nmid 3$ due to $3 \equiv 3 \pmod{8}$.

Hence, 2 is prime.

(2) $\alpha = 1 \pm \sqrt{3}$ is prime.

Let $\beta | \alpha$. Then $|N(\beta)| |N(\alpha)| = 4$. So $|N(\beta)| = 1, 2, 4$.

The previous argument shows that β is either a unit or

an associate of α . So α is prime.

$$2. \mathbb{Z}[\sqrt{-5}] \ (d=-5): \ 6 = 2 \cdot 3 = (1+\sqrt{-5})(1-\sqrt{-5}).$$

Ex. Show $2, 3, 1 \pm \sqrt{-5}$ are prime in $\mathbb{Z}[\sqrt{-5}]$.

$$3. \mathbb{Z}[\sqrt{5}] \ (d=5): \ 4 = 2 \cdot 2 = (3+\sqrt{5})(3-\sqrt{5}).$$

Will show next time that $2, 3 \pm \sqrt{5}$ are prime.